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MATE CHOICE BASED UPON NATURALLY OCCURRING COLOR-PATTERN VARIATION IN A GUPPY POPULATION

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Abstract.—This study tested the effect of differences in the extent of orange pigment in the color pattern of male guppies on the sexual responsiveness of females. Fish used in this study were descendants of a single natural population from the Paria River of Trinidad. Males from this population have unusually large, brilliant orange spots. I used three experimental approaches to test for discrimination by females among males based on the relative area of orange in color patterns: 1) the time to mating when a male was presented to a virgin female; 2) the frequency of sexual responses of females to passing, nondisplaying males; and 3) the proportion of a male's courtship displays that elicited a female sexual response. In all three experiments, females appeared to discriminate against males with less-than-average amounts of orange in their color patterns. In at least one experiment, however, the increase in female responsiveness with increasing amounts of orange leveled off and possibly decreased at high levels of orangeness. This suggests that there may be no advantage of increased amounts of orange above a certain level. These results suggest that female choice is a mechanism for the evolution of color patterns in guppies and may have contributed to the distinctive color pattern of the Paria population.

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Darwin (1871) argued that female mating preferences could explain the evolution of the conspicuous sexual display characters of males of many species. This hypothesis is of particular interest because it can explain the evolution of exaggerated, seemingly costly traits such as the elaborate plumes of peacocks and birds-of-paradise.

Female choice has often been inferred from patterns of male mating success with only negative evidence to rule out other factors such as male-male competition that might affect mating success (e.g., Andersson, 1982a). Darwin (1871) discussed competition among males for access to females as a mechanism of sexual selection in addition to female choice. Since both can affect male mating success, data on mating success alone are not sufficient to demonstrate female choice.

In order to demonstrate that female choice affects the evolution of a male sexual character, it is necessary to show that variation in the male character affects female sexual behavior and that the differences in female behavior lead to differential mating success in males. Then, if the variation in the male character is heritable, differential mating success (i.e., sexual selection) can result in evolutionary change. A lack of difference in female behavior does not rule out the possibility that choice is occurring, however; see Partridge and Halliday (1984).

There has been relatively little direct documentation of female choice, and the data are largely confined to species in which females can obtain direct fitness benefits such as access to a high-quality territory (Pleszczynska, 1978), food (Thornhill, 1980), or protection (Borgia, 1981). On the other

hand, theoretical models of sexual selection (e.g., Zahavi, 1975; Lande, 1981; Andersson, 1982*b*; Kirkpatrick, 1982, 1986; and many others) generally assume that the evolutionary benefits of mate choice stem from inherited effects on offspring fitness rather than on immediate, direct effects on female reproductive success. In part, this is because many of the most extreme sexual display characters of males seem unlikely to have any direct effect on female fitness. Female choice based on such characters has been documented adequately in only a few instances (e.g., Andersson, 1982*a*; Borgia, 1985; Ryan, 1985). Far more information on the nature of female choice is needed before the theoretical models can be tested.

In this paper, I report the results of an experimental study of female choice in the guppy (*Poecilia reticulata*), a small tropical fish with a promiscuous mating system. The conspicuous, polymorphic color pattern of male guppies is a classic example of a sexual character that is assumed to have evolved by female choice. Endler's (1978, 1980) studies have shown that guppy color patterns tend to have larger, more numerous spots in low-predation populations than in high-predation populations, and that the size and number of spots increases in populations where predation has been reduced or eliminated experimentally. This suggests that sexual selection favoring brighter or more conspicuous males acts in opposition to selection by predators. In order to test the hypothesis that female choice is a mechanism of sexual selection in the guppy, I investigated the effect of differences in male color patterns on the sexual responsiveness of females from a single natural population. The results of previous studies of guppy sexual behavior (Farr, 1980; Endler, 1983; Bischoff et al., 1985; Kodric-Brown, 1985) suggest that female guppies prefer conspicuous males, but no studies have examined the effect of within-population variation in male color patterns on female sexual behavior. Studying variation within a population is especially important in view of the variability in color patterns, life histories, genetic basis of color patterns, and behavior that has been documented among guppy populations (Haskins et al., 1961; Farr,

1975; Endler, 1978, 1980, 1983; Reznick and Endler, 1982).

I used three experimental approaches to test for differences in female sexual behavior toward different males: the time to mating of virgin females presented to single males, the frequency of sexual responses of virgin females to males that swam past but did not display, and the relative frequency of sexual responses of females to male courtship displays. Each of these measures of female responsiveness could have a direct effect on male mating success.

For these experiments, I used second- and third-generation descendants of guppies collected from a tributary of the Paria River, in the Northern Range of Trinidad, West Indies. The color pattern of male guppies from the Paria River is dominated by patches of unusually brilliant orange pigment, along with smaller areas of green, black, blue, yellow, and white. I chose to study the relative area of orange in the color pattern as a possible basis for female choice in Paria guppies because the orange pigment contributes to the conspicuous appearance of Paria males, and because of striking differences among males in the area of orange pigment. Preliminary data (Houde, unpubl.) based on the offspring of six males each mated to two females indicate that differences in the extent of orange in male color patterns have significant heritability.

MATERIALS AND METHODS

Guppy Biology and Sexual Behavior

Guppies are ovoviparous tropical fish, native to Trinidad and adjacent parts of South America. Males court females by performing a series of displays in which the body is vibrated stiffly in front of the female. Receptive females approach the male with a distinctive gliding motion, sometimes with the body arched toward the male. For the experiments described here, I judged that a female had responded to a male if she: 1) ceased her current activity; 2) moved unambiguously toward the male, following his movements; and 3) approached with a gliding motion rather than the normal jerky swimming motion. Further details of guppy courtship behavior are described by Baerends et al. (1955) and Liley (1966). During

a copulation, the male inserts his modified anal fin, the gonopodium, into the female's cloaca and transfers sperm. The embryos develop within the female for 3–4 weeks before birth. There is no parental care.

Guppies have a free-for-all promiscuous mating system, in which males approach, display and attempt to copulate with successive females. Males do not defend territories. Females are only receptive as virgins and just after giving birth to young, so only a small proportion of the females in a population are receptive at a given time. Males generally approach unattended females and tend to avoid displaying when other males are nearby (pers. observ.). In the field and in large aquaria, obvious male-male competition is uncommon and takes the form of chases, in which two males pursue a female, jockeying for position, rather than any form of overt aggression.

Origin and Maintenance of Experimental Fish

Guppies for the experiments were obtained by isolating pregnant females in 10-liter aquaria and raising their young. Males and females were separated at 4–6 weeks of age under MS-222 anaesthesia, and they were raised to maturity in 40-liter aquaria. All fish were fed twice daily, with liver paste in the morning and newly hatched brine shrimp nauplii in the afternoon. The room in which fish were housed was maintained at 25–27°C on a 12-hr light, 12-hr dark photoperiod. Local "dawn" came at 7:00 A.M.

I used males that were 3–5 months old for the experiments. Males in each experiment were sons of different females isolated from stock aquaria, and so were related at most as paternal half-sibs. I chose virgin females at about 15 mm standard length (the minimum size of fully mature and colored males) because size, rather than age, appeared to be the best predictor of sexual maturity.

All experimental trials were conducted between 7:00 A.M. and 10:00 A.M., before feeding. Guppies in the laboratory are sexually most active and most consistent in their behavior at this time.

In order to quantify the amount of orange in male color patterns, I placed each fish in

a narrow viewing chamber and recorded the size, shape, position, and color of each element of the color pattern on standard guppy outline diagrams. I then calculated the area of each color spot relative to total body area (in side view) using the digitizer board of an HP-85 computer.

Experiment 1: Mating Speed

This experiment used mating speed, the length of courtship required for a male to obtain a copulation, as a measure of the sexual attractiveness of males. To determine the effect of the relative area of orange in the color pattern on mating speed, males were categorized into four classes of orangeness, ranging from the most orange "A" males to the least orange "D" males (mean percentage of body area that was orange for each class: A) 22%, B) 13%, C) 9%, D) 4%). For each experimental trial, four males matched to within two weeks in age were chosen, one from each of the orangeness classes, and were placed at random in separate aquaria.

Each male was allowed to acclimatize for 24 hr in his aquarium with a female. On the day of the trial, the resident female was removed from each aquarium, and a virgin female was introduced to each male in turn. The four females used for a trial were litter-mate sisters matched for size. This minimized variability in mating speed caused by differences in age or size or by genetic differences among females. For each male, the time from the first male display to copulation and the number of displays before copulation were recorded. If no mating occurred within ten minutes of the first display, observations were discontinued. This procedure was repeated on different days for a total of ten trials, each involving a different set of four males, for a total of 40 observations of mating speed.

Each aquarium for this experiment was maintained at 25°C with an electric heater. The aquaria contained fine, light-colored gravel. To control the color of the background further, tan paper was attached to the back and sides of each aquarium. The aquaria were illuminated by four 40-watt fluorescent ceiling lights mounted behind plastic diffuser panels.

Separate statistical analyses were carried out for two measures of mating speed: time from first display to mating and number of male displays before mating. I used Friedman's test for randomized blocks to test for differences in mating speed among orangeness classes (BMDP program P3S; Dixon and Brown, 1979). A nonparametric test was necessary because some males failed to mate. These males were ranked slowest but could not be assigned an actual mating speed. I used Page's test for ordered alternatives (Sokal and Rohlf, 1981 p. 547) to test for a trend of decreasing mating speed with increasing orangeness.

Experiment 2: Responses to Nondisplaying Males

This experiment took advantage of the observation that virgin females respond sexually to males that swim past without displaying. This behavior, which I call "move-toward" behavior, is otherwise identical to a sexual response to an actual male display (see criteria described above) and is directed only to males. Because move-toward behaviors occur independent of male displays, the relative frequency of move-toward behaviors elicited by a male should be a good measure of his attractiveness to females, with minimal confounding by male-male competition and differences in courting ability.

Observations of move-toward behaviors were conducted using experimental groupings consisting of six males chosen to vary in orangeness, and six large nonvirgin females. For each trial, a single virgin female, smaller than the resident nonvirgins, was introduced into the aquarium, was observed for 15 minutes or until a copulation occurred, and then was removed. The additional nonvirgin females were included in order to minimize competition among males. Move-toward behaviors performed by the virgin females were recorded, noting which male elicited the response in each case. Ten virgin females were observed with each of five different six-male groups (50 females total).

The experimental groups were housed in 200-liter aquaria. Each aquarium contained coarse, natural color gravel, several oyster

shells to control acidity, and a plastic partition that partly divided the aquarium and served to increase its effective size. As in the previous experiment, the back and sides of each aquarium were covered with tan paper. Each aquarium was illuminated with a single 15-watt "daylight" bulb 30 cm above the water surface and was maintained at 22–27°C by controlling the room air temperature.

Experiment 3: Responses to Displaying Males

This experiment tested for an effect of the extent of orange in the male color pattern on the proportion of male displays that elicit a female sexual response. The initial female response is the necessary first step in the sequence of behaviors leading to copulation, and so is likely to be directly related to mating success.

Female responses to displays (see criteria described above) were observed in experimental groupings consisting of four males chosen to vary in the extent of orange in the color pattern, and ten females recently introduced to the tank as virgins. Upon introduction to males, most naive virgin females copulate within a few minutes. Males often cease all courting activity for up to one hour after copulating, so observations immediately after introduction of females were not feasible. Females remain responsive to male displays for several days after introduction to males (see Results below), and copulations during this period are frequent enough to suggest that most females copulate at least once or twice after the first mating. Therefore, the observations for this experiment were conducted on the two days following the introduction of virgin females. Each observation session consisted of a ten-minute focal observation period for each of the four males in an experimental group, repeated the following day, for a total of 20 minutes of observations for each male in the experiment. For each male, the total number of courtship displays performed to females in the aquarium and the number of displays that elicited a female response were recorded. Eight different four-male groups (a total of 32 males) were observed in this way, each on the two consecutive days fol-

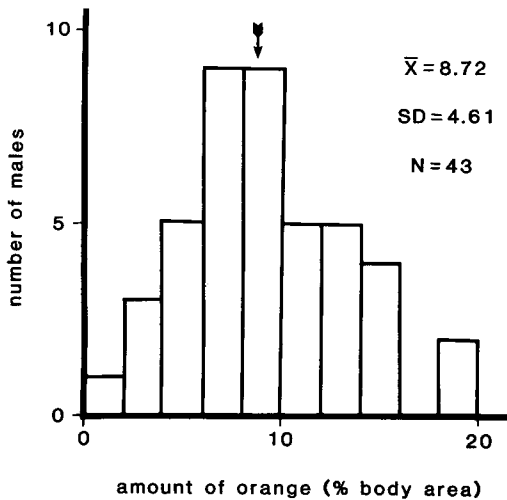


FIG. 1. Frequency histogram of relative area of orange in color patterns of wild-caught and lab-reared Paria males. Area of orange is expressed as percentage of total body area, including tail and dorsal fins. Arrow indicates sample mean.

lowing the introduction of virgin females. The experimental groups were housed in the same 200-liter aquaria as described for the previous experiment.

I used analysis of covariance (SAS statistical package, PROC GLM; SAS Institute Inc., 1982) to test for an effect of the extent of orange in a male's color pattern on female responsiveness in experiments 2 and 3 while controlling for any possible differences among replicate groups.

RESULTS

Male guppies from the Paria population varied greatly in the amount of orange in their color patterns. Orangeness ranged from no orange at all to orange pigment over nearly 30% of body area (in side view) for one experimental male. Figure 1 shows a frequency distribution of orangeness for a random sample of 28 wild-caught and 15 lab-reared Paria males. The mean orangeness for this sample was 8.7% of total body area. Wild-caught and lab-reared males did not differ significantly in orangeness ($t = 1.12$, $d.f. = 41$, $P > 0.20$).

In all three experiments, female guppies responded most strongly to males with more-than-average amounts of orange in their color pattern.

TABLE 1. Ranked data on mating speed of males from four orangeness classes (A: most orange, D: least orange) in experiment 1. Each line of data represents a different trial, employing a different set of four males. Data are ranked within trials (1: fastest, 4: slowest). Dashes indicate males that failed to mate (ranked slowest). Time to mating = time from first display to mating. Displays to mating = number of displays before mating.

Trial number	Time to mating				Displays to mating			
	A	B	C	D	A	B	C	D
1	1	2	3	4	1	3	2	4
2	1	2	3	4	1	2	3	4
3	1	—	2	3	1.5	—	1.5	3
4	1	—	—	—	1	—	—	—
5	1	—	2	3	1	—	2	3
6	2	3	1	—	2	3	1	—
7	—	1	—	—	—	1	—	—
8	1	2	—	—	1	2	—	—
9	1	2	—	—	1.5	1.5	—	—
10	1	2	—	—	1	2	—	—

Experiment 1: Mating Speed

This experiment was designed to test the effect of the male's color pattern on the female's willingness to mate by measuring mating speed.

The rank orders of the two measures of mating speed, time from first display to copulation and number of displays before copulation, are shown in Table 1. Data were ranked within each trial. Females mated soonest with the male with the most orange in his color pattern (class A males) in eight of the ten trials. Females tended to be most reluctant to mate with the least orange male. Both measures of mating speed gave similar results. Friedman's tests confirm that there were significant differences in mating speed among the male-orangeness classes (time to mating $X^2 = 14.43$, $d.f. = 9$, $p = 0.0024$; displays before mating $X^2 = 12.75$, $d.f. = 9$, $P = 0.0052$). Page's test revealed a significant trend of decreasing mating speed with increasing orangeness across categories (time to mating: $L = 283.5$, $P < 0.001$; displays to mating: $L = 281.0$, $P < 0.001$).

Of the 40 males tested, 15 failed to mate within ten minutes. Males that failed to mate within ten minutes had significantly less orange in their color patterns than males that did mate ($t = 2.83$, $d.f. = 38$, $P = 0.0073$).

There were no differences among orangeness classes in the time from the introduc-

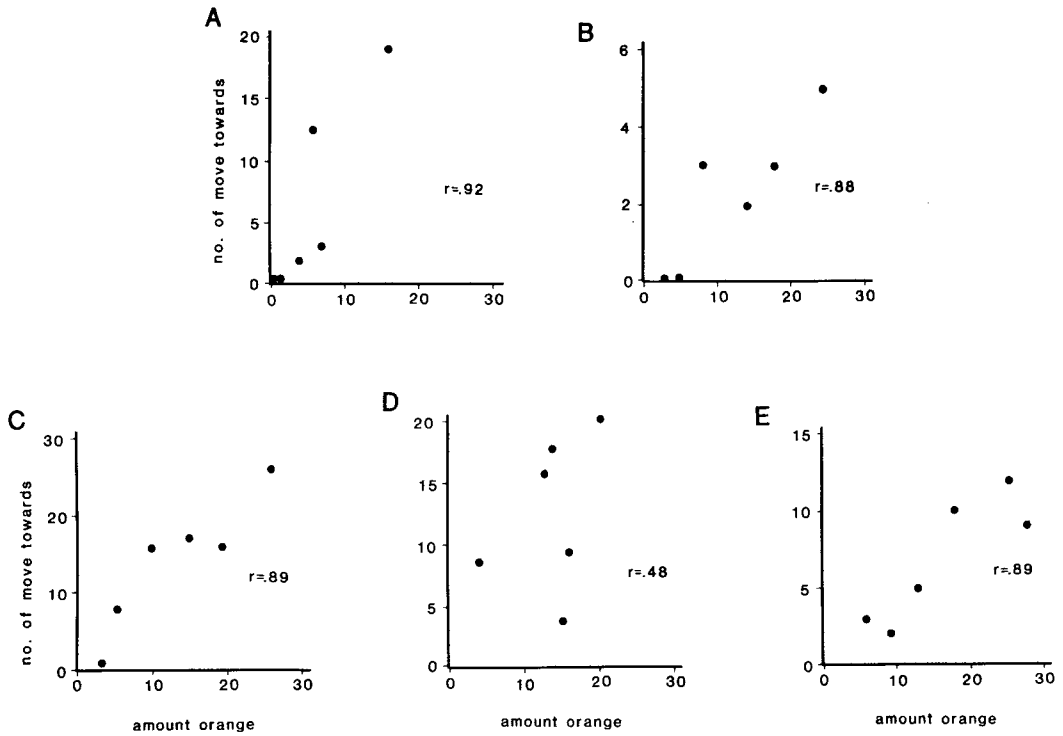


FIG. 2. Effect of amount of orange in male color pattern on number of move-toward behaviors received (experiment 2). Data are plotted separately for five replicate six-male groups. Correlation coefficient for each experimental group is shown.

tion of the female to the male's first display ($X_r^2 = 2.43$, $d.f. = 9$, $P = 0.4481$), or in rate of courtship (number of displays per minute: $X_r^2 = 1.32$, $d.f. = 9$, $P = 0.7244$). This supports the interpretation that female behavior was affected by the male color pattern rather than by correlated differences in sexual behavior.

Experiment 2: Responses to Nondisplaying Males

This experiment tested for differences in the frequency of sexual responses of females to passing nondisplaying males. Similar to the previous experiment, virgin females were most attracted to males with the most orange in their color patterns. In all five experimental groups, males with more orange received more move-toward behaviors than males with less orange. In Figure 2, the relationship of male orangeness to the number of move-toward behaviors he received is plotted for each experimental group of six

males. The scales on the vertical axes differ because move-toward behaviors were more common in some groups than in others. In each case, there was a strong positive correlation between orangeness and number of move-toward behaviors. I used analysis of covariance to test for an overall effect of orangeness while controlling for differences in the average number of move-toward behaviors among experimental groups. The effect of orangeness on number of move-toward behaviors was highly significant (Table 2).

Males in this experiment tended to ignore virgin females and appeared to prefer to court the larger nonvirgins, even though the virgins were more responsive sexually. Thus, move-toward behaviors occurred mainly when a male swam by the virgin female in pursuit of a nonvirgin, and so these behaviors were not affected by male-male competition for virgin females. Further, the swimming activity (and hence the potential

TABLE 2. Analyses of covariance for experiments 2 and 3 giving tests for a linear effect of extent of orange in the male color pattern on female sexual responsiveness (frequency of move-toward behaviors in experiment 2, number of responses/number of displays for experiment 3) while controlling for differences among experimental groups. *F*-tests are based on type I sums of squares.

	Effect	<i>F</i>	<i>d.f.</i>	<i>P</i>
Experiment 2:	Group	7.16	4,24	0.0006
	Orange-ness	21.69	1,24	0.0001
Experiment 3:	Group	2.04	7,23	0.0926
	Orange-ness	26.16	1,23	0.0001

for close encounters with virgin females) did not appear to differ among males. There was no relationship between display rate and orangeness ($F = 1.13$, $d.f. = 1,22$, $P = 0.2993$; no data from group 1), indicating that color pattern had no major effect on the ability of males to court females.

Experiment 3: Responses to Displaying Males

This experiment tested for differences in the responsiveness of females to male displays. The results, shown in Figure 3, indicate that the ability of males to elicit sexual responses of females generally increased with increasing area of orange in the color pattern, at least to a point (see below). Under these experimental conditions, females responded to as few as 6.6% of displays for one male with no orange in his color pattern to more than 50% of displays for males with large amounts of orange. Statistical analysis revealed a significant effect of male orangeness on the proportion of displays that elicited female responses when differences in responsiveness among experimental groups were controlled (Table 2). As in experiments 1 and 2, there was no relationship between orangeness and display rate, (ANCOVA: orangeness $F = 1.20$, $d.f. = 1,12$, $P = 0.2971$).

The data shown in Figures 2 and 3 can be interpreted as average female "preference curves" (Lande, 1981; Cohen, 1984; Heisler, 1984, 1985; Kirkpatrick, 1986) indicating the effect of variation in a male trait (e.g., extent of orange) on female prefer-

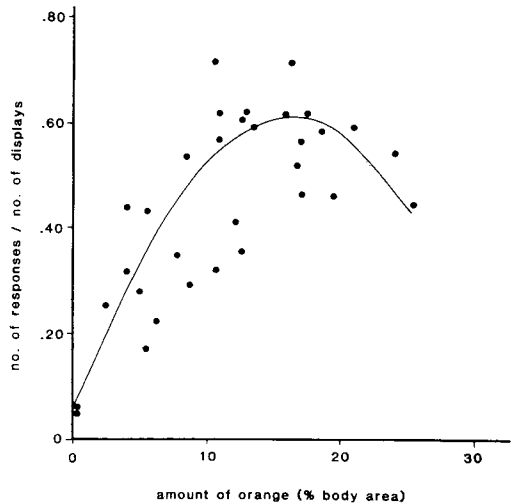


FIG. 3. Effect of the amount of orange in male color pattern on the proportion of displays eliciting female sexual responses (experiment 3). Each point represents one male. Data are based on observations of eight experimental groups of four males each. Best fit quadratic curve from analysis in Table 3 is plotted.

ences. The analyses of covariance presented above tested for a linear effect of orangeness, corresponding to "open-ended" preference curves, in which female preferences increase indefinitely (but linearly in this case) with increasing male trait values. The data from experiment 3 (Fig. 3), in particular, suggest a nonlinear relationship, in which the average preference curve levels off and possibly decreases at high values of orangeness.

To test this, I incorporated a quadratic term into the statistical models. Table 3 gives the ANCOVA results based on type I sums of squares for the data sets from experiments 2 and 3. *F*-statistics based on type I sums of squares test whether each term makes a significant additional contribution to the fit of the model once previously listed terms have been incorporated (Freund and Littell, 1981). R^2 values for the linear and quadratic models are compared in Table 4. For both experiments, the fitted quadratic model corresponds to a preference curve that levels off and, in experiment 3, possibly declines at high values of orangeness (see Fig. 3). Adding the quadratic term significantly improves the fit of the model for experiment 3 (Table 3) and results in a large increase in

TABLE 3. Analyses of covariance as in Table 2, with the addition of a quadratic term for extent of orange (see text).

Effect		F	df.	P
Experiment 2:	Group	7.67	4,23	0.0004
	Orangeness (linear)	23.25	1,23	0.0001
	Orangeness (quadratic)	2.73	1,23	0.1123
Experiment 3:	Group	6.12	7,23	0.0005
	Orangeness (linear)	78.36	1,22	0.0001
	Orangeness (quadratic)	46.89	1,22	0.0001

R^2 (Table 4). The quadratic term is significant even when data points for the males with the two highest and the two lowest orangeness values are deleted. In experiment 2, adding the quadratic term resulted in only a small increase in R^2 , but the significance level ($P = 0.11$) is low enough to be suggestive (combined probability for significance of the quadratic term in the two experiments: $P < 0.001$). These results suggest that the average preference curve of female guppies increases with male orangeness to a point, but levels off or possibly decreases at high orangeness values.

DISCUSSION

Guppies from the Paria River and adjacent streams flowing north from Trinidad's Northern Range are distinct from other guppy populations in their large, brilliant orange spots (Endler, 1978, 1983). The experiments described here demonstrate that female guppies from the Paria population have an active sexual preference based on variation in the amount of orange in the color patterns of males. These results suggest that female choice plays an important role in sexual selection and the evolution of color patterns in guppies. Further, the particular pattern of preferences based on orangeness may be responsible for the distinctively orange color pattern of Paria guppies. Endler (1978) has suggested that the brilliant orange color may be conspicuous to female guppies but inconspicuous to the probable major predator in these populations, *Macrobrachium*, a prawn.

I argue that female guppies in these experiments displayed "active" preferences (sensu Partridge and Halliday, 1984) because they behaved differently towards dif-

ferent males, depending on the amount of orange in the color pattern. Females varied in their willingness to copulate (experiment 1) and in sexual responsiveness to males (experiments 2 and 3). There may have been an element of "passive" choice (sensu Parker, 1983), especially in experiment 2 (responses to nondisplaying males), in that males with more orange could have been more conspicuous to females, attracting them from a greater distance. The distance from which passing males elicited move-toward behaviors (about 3 cm) appeared similar for all males, however.

A major difficulty in documenting female choice has been in distinguishing the effects of male-male competition from those of female choice (e.g., Borgia, 1981; Partridge and Halliday, 1984). Both can affect male mating success. Since there was no relationship of male orangeness to courtship behavior in any of the three experiments, it is unlikely that male-male competition was responsible for the observed differences in responsiveness of females. Sexual competition among males was clearly not a factor in experiment 1, in which males were tested individually in separate aquaria. In experiment 2, females responded to nondisplaying males, so the results were unlikely to have been affected by competition among

TABLE 4. Comparison of relative amounts of variation in female responsiveness explained by the linear and quadratic statistical models shown in Tables 2 and 3.

	R^2	
	Linear	Quadratic
Experiment 2:	0.68	0.71
Experiment 3:	0.64	0.88

males attempting to monopolize females. Male-male competition was potentially more important in experiment 3, in which females responded to displaying males, yet the results were similar to those of the other two experiments. Indeed, two males attempted to court the same female only rarely in experiment 3, probably due to the excess of females in the experimental groupings, so that male displays almost always occurred away from other males.

Each of the three experiments tested for an effect of orangeness on female behavior at a different stage of courtship. Experiment 1 tested the effect of male orangeness on the time to completion of the entire sequence of courtship. Mating speed ultimately depends on the female's willingness to copulate, but it could also be affected by her initial responsiveness to male displays. Experiment 2 tested the effect of orangeness on the sexual attractiveness of males independent of their displays. Experiment 3 tested the effect of orangeness on male attractiveness during displays.

The fact that all three experiments demonstrated a preference based on the extent of orange in the male color pattern indicates that: 1) differences in time to mating (experiment 1), a measure of a male's ability to obtain copulations, are probably in part due to differences in the male's ability to elicit sexual responses (experiment 3); 2) male display is not critical to the expression of female preferences (experiment 2 vs. experiment 3); and 3) variation in the opportunity for competition among males (none in experiment 1, higher in experiment 3) had little effect on the observed preference patterns.

Taken together, the results of previous studies of mate choice in guppies and the results presented here indicate a general pattern of female preferences for conspicuous males, and in particular for males with carotenoid (red-orange-yellow) spots in their color patterns. Female guppies appear to prefer males with more total spots (Endler, 1983), domesticated "fancy" strains over wild strains (Kodric-Brown, 1985), and longer-tailed domesticated strains (Bischoff et al., 1985). These studies suggest a general preference for conspicuous males. Endler's

(1983) experiments on the effect of background gravel colors showed that males with colors similar to the background were at a disadvantage relative to males with contrasting colors. This is strong evidence that relative conspicuousness is important in female choice. On the other hand, Endler (1983) found a significant preference for males with more carotenoid colors, once background effects were controlled. Similarly, Kodric-Brown (1985) found that carotenoid colors had a greater effect on female preferences than did other colors. The results of the experiments described here confirm that the relative area of orange in the male color pattern has an effect on the sexual behavior of female guppies. Since the extent of orange probably affects the general conspicuousness of males, however, it is not clear whether the preferences of Paria females are based on the conspicuousness of males, or specifically on orangeness.

If responsiveness to displays affects male mating success directly, the results of experiment 3 imply that there is a net directional effect of female choice. The most attractive males were those with about 16% orange (Fig. 3), while average orangeness was only 9% (Fig. 1). Countervailing selection (e.g., by predators) would be necessary to maintain an equilibrium. The leveling-off of the attractiveness of males in experiment 3 suggests that there may be no selective advantage through female choice to males with more than about 16% orange. The leveling-off of the average preference curve also suggests that the preference curves of individual females are not open-ended. Nevertheless, large evolutionary changes in orangeness could occur if genetic variation in female choice allowed evolutionary change in the average preference in the population.

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